

DISCUSSION PAPER

The past, present and future of nursing education in the People's Republic of China: a discussion paper

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Abstract

Aim. This article presents a discussion of nursing education development in the People's Republic of China in its historical, economic and sociopolitical contexts.

Background. China has a population of 1.3 billion with about 2.18 million nurses. With the recent surging economic and social development in China, nursing education has undergone transformation changes in the past two decades.

Data sources. Online bibliographical databases from 1990 to 2010 were searched including CINAHL, MEDLINE, Wan Fang Data and Chinese National Knowledge Infrastructure. Search terms included nursing education, China and development.

Methods. Thematic analysis and narrative synthesis were used to identify and report themes from literature.

Results. Database searches yielded 1674 papers, and 34 met the inclusion criteria for review. The standard of nursing education varies greatly in different parts of China, because of its huge size and population, with pre-registration programmes offered at the secondary, associate degree and baccalaureate level. Multi-level nursing education is one of the major barriers for professional development. There is a need to upgrade the pre-registration education to at least associate degree level. There is also a need to enhance graduate nursing education at master and doctoral level to prepare advanced practice nurses, nurse scientists and nursing faculty.

Conclusion. The challenges for nursing education development in China are echoed and encountered in many parts of the world. The experience in China and the lessons learned would be relevant to developing countries. Nursing in China must continue to develop in parallel to international trends. Promoting communication and maintaining international links are important for the global development of nursing practice.

Keywords: baccalaureate nursing programmes, China, doctoral nursing programmes, nursing education

Introduction

Mainland China, with its fast economic growth, is rising to be an influential country to the world economy and power.

The People's Republic of China (PRC) has a population of 1.3 billion with about 2.18 million nurses (Guo 2011). The Chinese Nursing Association had its centennial celebration in August 2009. It is timely to review nursing education

development in the PRC for the past decades and plan future direction.

Background

Modern nursing practice and nurse education in China dates back to as early as 1888 when the first British nurse came to the Chinese kingdom of the Qing dynasty. In the early 1900s, academic nursing programmes were offered at both the secondary and baccalaureate levels with influence from both the United Kingdom and the United States of America (USA). Such a multi-level nursing education system had gone through the world war and civil war, and continued after the establishment of the PRC.

In 1979, the PRC started the huge economic reform, and the country has been open to the outside world since then. With increased interactions with the rest of the world, the PRC has been affected directly by the economic globalization. The healthcare needs have changed with the fast occurrence in industrialization and urbanization. The population of people from other countries coming to work and stay in the PRC is growing fast (Watson 2005), adding to the challenges to the healthcare system. There has also been increasing complexity of needs of clients from diverse cultural and socioeconomic backgrounds, and increasing demand for quality care and effective use of the finite resources. Nursing in the PRC is therefore facing similar challenges to other countries in the world. Nurses need to be multi-skilled with clinical expertise, research competence, information technology skills, management ability and cultural competence. Nurses, as key members in the healthcare team, require a much higher level of professional competence than in the past.

The aim of this article was to discuss the nursing education development in the PRC and to examine the characteristics of its current development. Insights are offered to other countries in nursing and nursing education development.

Data sources

Search methods

A search of electric databases was supplemented with a manual search of references in relevant studies for the period from 1990 to August 2010. The following electronic databases were used: CINAHL 1990–2010, MEDLINE 1990–2010, WanFang Data (In Chinese, 1990–2010) and Chinese National Knowledge Infrastructure (CNKI, in Chinese, 1990–2000). Search terms included (Nursing education) AND (China) AND (Development) AND (Nursing Practice Development). The search was limited to publications since

1990 to review publications that might have more relevance to recent nursing development.

Papers that met the following criteria were reviewed: (1) focusing on nursing education development in PRC, (2) the language has to be Chinese or English, (3) the journal is peer-reviewed and (4) papers published from 1990 onwards. Papers were excluded if their focus was only limited to one hospital, one school or one city in mainland China.

The references and abstracts identified from the search were assessed against the inclusion/exclusion criteria independently by two reviewers and full text were obtained for relevant studies. Decisions to include a publication were made independently by the two reviewers following evaluation of the full text of all retrieved papers. Disagreements were resolved by consensus with close attention to the inclusion/exclusion criteria.

Search outcome and quality appraisal

A total of 1674 papers were retrieved: 1030 from CNKI, 338 from WanFang Data, 154 from CINAL and 152 from Medline. The title and abstracts were reviewed. Articles were discarded if they were duplicated across databases or if they did not meet the inclusion criteria. There is no gold standard for evaluating and interpreting quality in literature reviews. We did not exclude any papers on evidence strength (Whittemore & Knalf 2005).

Thirty-two papers and two book chapters meeting the inclusion criteria were identified. Thus, a total of 34 papers (nine surveys, 14 discussion paper, four reviews, four proceedings, two book chapters and one newsletter) were reviewed with nine in the English language and 25 in the Chinese language (Figure 1).

The Chinese literature was written by nursing academics working in universities in the Mainland China and the majority have a doctoral qualification. There was no need to translate Chinese paper to English in this review, as all the authors are proficient in both the Chinese and the English languages.

Data synthesis and results

Thematic analysis, a common technique used in the analysis of qualitative data in primary research, can be used to systematically identify the main, recurrent and/or most important themes (based on the review question) across multiple evidences (Popay *et al.* 2006). We chose thematic analysis from the toolbox in Popay *et al.*'s (2006) report as an appropriate approach to analyse and understand evidence from included studies, which all focus on nursing education development in PRC. Popay *et al.*'s (2006) method of combining narratives allowed the synthesis to be done in a

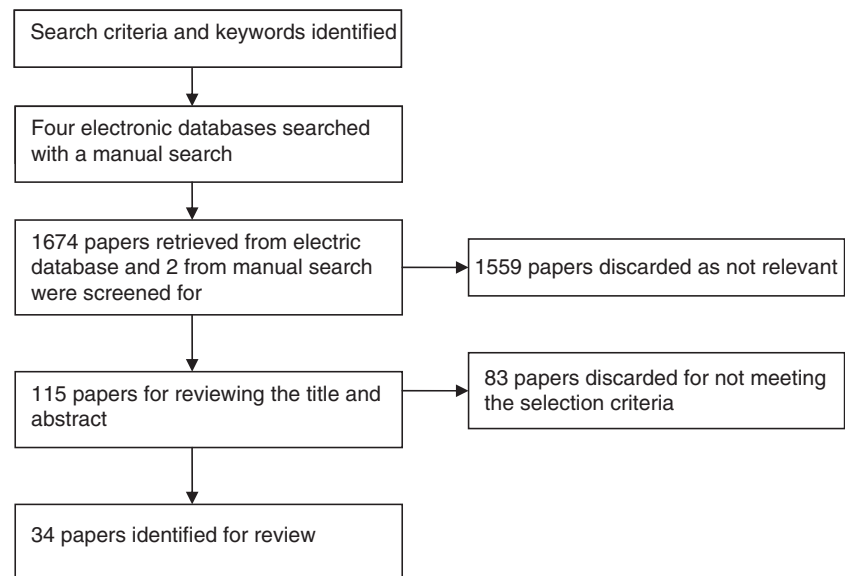


Figure 1 Literature review flowchart.

rigorous and methodical fashion. Findings were then reported in a narrative way. Each paper was reviewed and paper-based thematic analysis was conducted by the first author, with the other authors concurring with the themes after reading the included papers.

Eight themes were identified from the review: (1) the early development, (2) nursing regulation, (3) nursing education at the secondary level, (4) nursing programmes at the university diploma level, (5) baccalaureate nursing programmes, (6) master of nursing programmes, (7) doctoral nursing programmes and (8) evolving nursing education and practice.

Discussion

The early development

In 1888, modern nurse training in China began with a 3-year hospital-based programme in Fuzhou when Western missionaries introduced modern nursing (Chen 1996, Chan & Wong 1999). Nursing education at the tertiary level began in 1920 when Peking Union Medical College (PUMC) established the first baccalaureate nursing programme with a 5-year curriculum. The programme was funded by Rockefeller Foundation of the USA (Dai 1990, Chen 1996). This baccalaureate nursing programme remained one of its kind in China until 1952 when a nationwide restructuring of the higher education system was launched. As a result of the restructuring, the PUMC School of Nursing was downgraded to a vocation training institute, and the baccalaureate nursing programme was replaced by a vocational nurse training. During the period from 1924 to 1952, a total of 264 nurses graduated from the PUMC baccalaureate programme (Dai

1990). Many of them became nurse leaders with significant contributions to nursing professional development in China.

During the period of the Cultural Revolution in the PRC from 1967 to 1977, most tertiary education activities were suspended. All nursing schools were abolished, leaving some 'health training programmes' operated at the junior and senior high school levels. Graduates from these programmes were assigned to take up the nursing work. Nursing education and professional activities came to a standstill at that time (Chan & Wong 1999, Xu *et al.* 2000). When the Cultural Revolution ended in 1977, formal education was gradually restored nationally, and so was nursing education.

The year 1979 marked the beginning of the enormous economic reforms in the PRC with opening of the country to the outside world. The following few decades have seen the PRC develop itself to a huge economic entity with drastic social changes. Nursing education in the PRC also started to progress, though in a speed not as fast as the country's economic development. Many universities in the big cities started to offer their own baccalaureate nursing programmes. At the same time many post-basic nursing programmes were established for the secondary-level nurses to obtain higher academic qualifications. Postgraduate nursing programmes at the masters and doctoral levels appeared in the late 1990s. Today's nursing education in the PRC is characterized by a multi-level multi-mode system (Xu *et al.* 2000).

Nursing regulation

In 1909, the CNA was established, and started administering examinations for certification of nurses in 1914. However, it is not known whether such certification examination was

applicable to all nurses in the country, and when examinations had been suspended (Wang & Zhao 2009). In 1993, the 'Nurses Regulation Method' was established, which was an ordinance to regulate nursing practice by the PRC Ministry of Health (MoH) (MoH 1993). All individuals required the 'Nursing Practice Certificate' for the nurse registration. The MoH held an annual National Nurse Licensing Examination (NNLE). Those who pass NNLE will be issued the Nursing Practice Certificate. Graduates from all tertiary nursing programmes, including diploma, associate degree or bachelor degree levels, were exempted from the NNLE. Only graduates of those non-exempted secondary nursing programmes need to sit for the NNLE.

The nurse regulation system was modified in 2008 when the first 'Nurses Act' in the country was passed. Nursing registration remains localized at provincial level, but nurses could work everywhere in the country through notification by the registration bodies that are the health offices at Provincial level. The registration lasts for 5 years. The Nurses Act specifies that all basic nursing programmes must be accredited by the Ministry of Education or the MoH. They must be full-time study of at least 3 years in length, and must include clinical placement for at least 8 months. Nurses who graduate from the basic programmes are considered generalists and eligible to practice in all types of clinical services. Furthermore, no exemption for the NNLE is given to any type of basic nursing programmes.

Nursing education at the secondary level

The Ministry of Education classifies the nursing programmes at the secondary level as 'secondary vocational education'. These programmes offer skill-based nursing in a 'Health School' which may be an independent institute, an operational unit affiliated to a hospital or an academic unit attached to a medical university/college (Huang & Li 2006).

There are two types of secondary nursing programmes. For the first type, the programme enrolls high school graduates who have completed 6 years of secondary education. The curriculum of these programmes could be 2 or 3 year in length. For the second type, the programme admits middle or junior high school graduates, and has a 3- or 4-year curriculum. The students enrolled to this programme only have 3-year secondary education and are at the age of around 15 years. Graduates from such programme are similar to other high school leavers except that they have had nursing training. Nursing programmes of this second type are still producing the majority of nurses in the PRC. Most graduates from these programmes are around the age of 18 years. As a result of the lower level of academic background and the

young age of the students, there has always been a concern if the nurses trained in this type of programme are adequately prepared for the increasingly demanding role of a professional nurse.

For many years, there had been discussion that these secondary nursing programmes should be discontinued (Chan & Wong 1999). A consensus was reached in the nursing community in the late 1990s that the secondary nurse training would be phased out gradually (Xu *et al.* 2000, Sherwood & Liu 2005). There are currently 862 secondary nursing programmes nationwide. A recent survey showed that the enrolment of the secondary nursing programmes in 11 provinces had decreased from 83.47% in 1998 to 60.03% in 2007, whereas the enrolment of the university diploma programmes had increased from 13.14% in 1998 to 30.85% in 2007 and the enrolment of the baccalaureate programmes had increased from 3.39% to 9.12% in 2007 (You *et al.* 2010). However, as the capacity of nursing programmes at all levels in the PRC has expanded sharply in the past decade, the actual number of students enrolled to the secondary nursing programmes is larger than before.

Nursing programmes at the university diploma level

The university nursing diploma in the PRC is 3-year full-time programme. This is generally considered to be equivalent to the associated degree programmes in the community colleges in Western countries. This programme is to prepare nurses with better professional knowledge and clinical competence than those from the secondary nursing programmes (Huang & Li 2006). The number of diploma programmes increased sharply in the past decade from 40 in 1996 to 307 in 2007 in the whole country (You *et al.* 2010). Around 70,000 nurses graduated from the diploma programmes each year (Ren & Zhu 2007).

Baccalaureate nursing programmes

The baccalaureate nursing programme that had been stopped since 1952 was reinstated in 1983 at Tianjin Medical College (currently Tianjin Medical University) with 30 students in the first intake (Ma 2009). Since then, baccalaureate nursing education in the PRC has been expanding rapidly.

In 2005, the MoH issued the first 5-year plan on nursing development titled 'Outline of the Development Plan of Nursing Services in P.R. China 2005–2010' (MoH 2005). This plan addressed the issues of higher demand for quality health care, advancement of technology and evolution of delivery mode of health services. It then identified four areas of nursing education that need to be addressed. These areas

are: the multi-level structure of the nursing education system needs to be modified and improved; the quality of nursing education needs to be promoted; post-basic nursing education and continuing nursing education need to be re-structured and regulated; and nursing education programmes must be designed for meeting the service needs, especially the need of specialty nursing practice and management and leadership in nursing (MoH 2005). As the MoH policy statement was made, the development of baccalaureate nursing programmes has accelerated further. In 2006, there were a total of 192 baccalaureate nursing programmes with more than 20,000 students in the country (Zheng 2009).

Although there are increasing numbers of universities offering baccalaureate nursing programmes, difficulty in recruiting students to nursing programmes is a common problem experienced by universities. The baccalaureate nursing programmes have to compete with degree programmes of other disciplines for recruiting high school graduates with better grades in the National University Admission Examination (NUAE). In many parts of the PRC, nursing is still perceived, in both economic and social terms, as a less desirable career. Nursing is not always the first choice of high school graduates with good NUAE grades. It is not uncommon that some students undertake the nursing programme because their NUAE grades were not good enough for a place in their preferred programmes. For these students, motivation to become a professional nurse is not strong (An *et al.* 2008).

Baccalaureate nursing programmes in the PRC can be of 5-year or 4-year duration, with the latter being more prevalent (Jiang 2007). The first 3-year curriculum usually consists of theoretical studies and then followed by 1 or 2 years of clinical practicum (Xu *et al.* 2002). The influence of the disease-centred medical education model on the nursing curriculum remains strong. This could be related to the fact that most nursing departments in the PRC universities are placed under the medical faculty or college. The curriculum of the nursing programmes mirrored that of medicine. Much weight is given to studies of the aetiology, pathology, diagnosis and treatment of diseases (Zhu *et al.* 2006). In the early years, many specialty nursing courses were taught by physicians. This was related to the lack of nurse educators with specialized qualifications (Xu *et al.* 2000). This situation has changed slightly in the past decade with gradual addition of humanities and social sciences courses to the curriculum and increasing number of nursing faculty with specialty qualifications. It is generally agreed in the nursing community that a holistic and health-centred nursing curriculum is way to go (Jiang 2007).

Master of nursing programmes

The first master degree nursing programme in the PRC was established at the Department of Nursing, Beijing Medical University (currently the School of Nursing, Beijing University) in 1990. Over the past 20 years, the master degree nursing programme has expanded from five programmes in 1998 to 65 in 2008 (Chang 1999, An *et al.* 2008, Li & Han 2010). Most of these programmes are 2 or 3 years in length (Shi 2008) which aim to prepare nurses to take up the role as nurse educators, researchers and managers (Wang *et al.* 2009).

One of the most significant developments of master level nursing education was the Programme on Higher Nursing Education Development (POHNED) (1993–2000) launched by the China Medical Board (CMB) of New York, which was evolved from the Rockefeller Foundation (Anders & Harrigan 2002, Sherwood & Liu 2005). The chief aim of this programme was to increase the number of master-prepared nursing faculty to meet the increasing need of baccalaureate nursing programmes. The CMB offered sponsorship for PRC nurses to study a 2-year master nursing programme that was developed in collaboration with the MoH of the PRC and Chiang Mai University in Thailand. This master nursing programme contributed significantly to channelling nursing education in the PRC away from medical dominance (Anders & Harrigan 2002, Sherwood & Liu 2005). The 2-year programme covered nursing courses in theory, research, education, clinical specialties and concepts central to Chinese culture, such as Traditional Chinese Medicine. The POHNED programme was housed in the nursing department at Xian Medical University in China. The programme was conducted in the English language and taught by faculty from the School of Nursing of Chiang Mai University. Students spent one semester at Chiang Mai University with a focus in community health and gerontology. Students are required to complete a thesis (Anders & Harrigan 2002, Sherwood & Liu 2005).

Between 1994 and 2000, five classes of the POHNED programme were completed with 84 nurses graduated (Anders & Harrigan 2002, Sherwood & Liu 2005). They formed a pool of better prepared faculty to teach on the baccalaureate nursing programmes and many of them become nurse leaders in the country. Although there are many other master nursing programmes offered in later days, none has produced such great impact compared with this milestone POHNED programme (Chan & Wong 1999).

Although the POHNED specifically aimed at preparing nursing faculty members, the other master nursing programmes that developed later were supposed to target at

advancing the professional competence. However, it was noted that many nurses in the PRC consider teaching in an academic institute a better choice than clinical practice (Hou *et al.* 2009, Li & Li 2009). Most master-prepared nurses chose to work in the academic rather than service sector, which is a phenomenon quite different from that in western countries [Institute of Medicine (IOM) 2011]. This could be a result of the huge demand for master-prepared faculty members in the universities, or, it could be because some master nursing programmes were not specifically designed to prepare the nurses to be good clinicians (Li 2006, Wang *et al.* 2009). Furthermore, there is not much difference in pay between academic and clinical settings, thus working in clinical setting might not be as attractive as working in academic setting.

Doctoral nursing programmes

Nursing education at the doctoral level is still at its infancy stage in the PRC. The first doctoral nursing programme was established in 2003 at Central South University and the Second Military Medical University. This first programme offered a reference for the later doctoral programmes in the PRC. By 2008, a total of ten doctoral nursing programmes were established in the country (Ma & Liu 2009). These are 3-year full-time or 3- to 7-year part-time programmes. The doctoral programmes aim to equip students with advanced competence in intellectual inquiry and research ability for generating nursing knowledge. Students have to complete course work and a research project. Most programmes require students to publish a paper in SCI or SSCI listed journals by the time of graduation. Graduates of the doctoral nursing programmes are expected to be nursing scholars (Ma & Liu 2009). As there are inadequate nurse faculty qualified to teach in doctoral programmes, a team approach is generally adopted for supervising the students. In most cases, some supervisors in the team are from disciplines other than nursing, such as medicine or from other universities (Ma & Liu 2009).

Evolving nursing education and practice

The development of nursing education in the PRC has been greatly influenced by the American model in the early stage and recent times (Xu *et al.* 2000). Today, in the PRC, there are about 862 secondary nursing programmes, 307 diploma programmes (You *et al.* 2010), 192 baccalaureate programmes (Zheng 2009), 65 masters programmes (Li & Han 2010) and 10 doctoral programmes (Ma & Liu 2009).

Like many developed countries, the PRC has an ageing population and increasing prevalence of chronic diseases.

There is strong need to prepare nurses who have the knowledge and skills in health promotion, disease prevention and management of chronic diseases [International Council of Nurses (ICN) 2007, Burman *et al.* 2009]. However, most pre-registration nursing programmes in China remain disease-oriented and focus primarily on acute care rather than community settings; and curricula are organized around traditional medical specialties such as medical-surgical or maternal-child. This situation is also common in many developed countries (Huang & Li 2006, IOM 2011). There is a need for re-examination and a radical revision of the pre-registration curriculum (Wang & Zhao 2009, Zheng 2009). To enable nursing graduates to meet future challenges in healthcare system, there is also need to develop a wide range of new competencies for nursing graduates such as disease prevention, health promotion, care-ordination, evidence-based practice, quality care and patient safety. Apart from acute care settings, students need clinical experience in community, long-term care and home settings, workplace or schools (IOM 2011).

The diverse levels of pre-registration nursing education in China continue to be a barrier to quality health care and nursing professional development. At present, it is unrealistic to expect a wholesale transfer of nursing education to degree level, as there is a need in the country for secondary and diploma level graduates especially at the rural areas (Ma 2010). Similar to many other countries, provision of university-level education for all pre-registration nursing programmes in China is a goal for the future. However, opportunities have to be provided for nurses to have educational mobility. Efforts to transition substantially more associate degree or diploma graduates into baccalaureate-level nursing education will be essential for developing the better-educated nursing workforce that the increasingly complex healthcare system requires. Working globally towards university-level education for professional nurses will require country-specific strategies that take into account national factors such as different entry points for education, cultural beliefs and norms, prior learning, experience and progression options (WHO 2009). It is the PRC government's plan to re-organize the nursing education system by expanding the baccalaureate programmes. The prime concern of the PRC government would be to meet the healthcare needs of people. The current multi-level system of nursing education would be maintained for some years. It is expected that more resources are required to upgrade all secondary nursing programmes to the tertiary sectors. Cost-effective studies could be conducted to see if the improvement of quality of nursing service justifies the additional resource put into the transfer of nursing education to the tertiary sector.

What is already known about this topic

- China has a population of 1.3 billion with about 2.18 million nurses.
- With the recent surging economic and social development in China, nursing education has undergone transformation changes in the past two decades.
- The majority of the review of nursing education development in China was published in Chinese, which could not be shared with international readers.

What this paper adds

- An in-depth analysis of the undergraduate and postgraduate nursing education development in China with particular focus on the past 20 years.
- Comparisons with an international nursing context.
- The challenges of contemporary nursing education development in China and their implications to global nursing development.

Implications for practice and/or policy

- The experience in nursing education development in China and the lessons learned would be relevant to developing countries.
- Promoting international links to build alliances and communicate ideas and best practices are important for the global development of nursing education and practice.

To improve patient outcomes currently and in the future, it is important that an evidence-based approach to nursing care be incorporated into clinical practice settings. A paradigm shift to evidence-based practice (EBP) has occurred throughout the nursing profession in western countries, such as Australia, Canada, the UK and Europe and the United States (Thompson 2004). Compared with western countries, evidence-based nursing lagged behind in Mainland China. EBP has not been regarded as an essential component of nursing practice, and introductory courses to EBP are rare and seldom mandatory in Mainland China (Jiao *et al.* 2009). Fortunately, a few international networking have been established, including the Chinese Cochrane centre, the Chinese branch of Joanna Briggs Institute-the FUDAN EBN centre (Hu & Li 2007). The international collaboration will accelerate the development of evidence-based nursing in Mainland China.

There has been increasing global recognition to the expertise of nurses practicing in specialty areas, for example

renal or diabetes care. These nurses have brought about significant improvement in patient outcomes (Byne *et al.* 2000, Williams & Jones 2006). Various positions or titles for these nurse experts have been established, for example, Advanced Practice Nurse (APN), Clinical Nurse Specialist (CNS), Nurse Practitioners (NP) or Advanced Practice Registered Nurse (APRN) (Daly & Carnwell 2003, Sheer & Wong 2008). Regulation and education systems have been established in some countries for these nurses. The International Council of Nurses (ICN) International Nurse Practitioner/Advanced Practice Nursing Network (2004) suggested that an ANP should be educated to the master degree. An advanced nursing practice curriculum is to prepare the practitioners to provide specialized care to clients with specific health problems, to conduct research and to coordinate and lead the specialized practice.

The concept of CNS was first introduced into the PRC in 2000s (You 2002). Education for clinical specialists began in the form of a certificate programme for enterostomal therapists. In the past decade, similar certificate programmes on acute care were conducted (Li *et al.* 2005). Contributions of the CNS to provision of cost-effective quality care have been recognized. In a survey of the physicians and nurses in the PRC, 99.7% ($n = 627$) of the respondents agreed that CNS should be developed in the PRC (Li *et al.* 2005). However, a regulation system for advanced practice is yet to be established. The current foci of master programmes in the PRC are on education and administration. It appears that a master degree in nursing specialty may be appropriate for preparing the nurse for advanced practice.

Some overseas countries, like USA, have developed Doctor of Nursing Practice (DNP) programmes to prepare graduates as clinical scholars who have the capacity to translate research, and to improve systems of care, patient outcomes, quality and safety. The DNP is the complement to other practice doctorates, such as MD or PharmD, that require highly rigorous clinical training (IOM 2011). This could be the way ahead for PRC to develop clinical nurse leaders.

Graduate nursing programmes will continue to develop in PRC (An *et al.* 2008, You *et al.* 2010). With the increased number of graduates in baccalaureate education, there will be greater pool of applicants for post-graduate programmes. Graduates from post-graduate programmes will be advance practice nurses, nurse scientists or nursing faculty whose responsibilities are to develop nursing and nursing knowledge. It is expected that more nurses will be equipped with higher academic qualifications in the future. Ultimately, it will benefit client care. In this way, nursing profession can be

on an upward trajectory (He & Guo 2007, Jiang 2007, An *et al.* 2008).

There is a worldwide nursing shortage and nurses are more mobile across countries. Nurses from PRC and prospective nursing students are being recruited to other countries. Policy makers, nurse administrators and nurse educators in PRC must work together to develop innovative short- and long-term solutions to retain nurses and attracting more people to join nursing. The public image of the nursing role must be enhanced to change outdated perceptions.

Conclusion

Given the huge size of the population in China and the nursing community, the professional development of nursing varies a lot in different parts of the country. The rapid economic growth, changing needs for healthcare, evolving healthcare system and increased influence from overseas, have all contributed to turning to a new page of the nursing education development in the PRC. The multi-level nursing education continues to be a barrier to nursing professional development. As the PRC has increasingly become a global player, nursing in the PRC must develop in parallel with international trends. The challenges of nursing education development in the PRC are echoed and encountered in many parts of the world. The experience in nursing education development in PRC and the lessons learned would also be relevant to developing countries. Promoting communication and maintaining international links are important for the future development of nursing practice and nursing education in the PRC.

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Conflict of interest

No conflict of interest has been declared by the authors.

Author contributions

L-LG was responsible for the study conception and design. L-LG performed the data collection. L-LG and CS performed the data analysis. L-LG, CS and BC were responsible for the drafting of the manuscript. L-LG, CS and BC made critical revisions to the paper for important intellectual content. CS provided administrative, technical or material support.

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